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Sayeon ERP

Project documentation and end-to-end business workflows

Audience	Developers, testers, administrators and business owners
Version	13 October 2026
Live site	https://sayeon.sanvima.net/
Part A	Project documentation – architecture, roles, data, operations
Part B	End-to-end business workflows – steps, approvals, statuses, outcomes

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Part A – Project documentation

A1. What the system is

A multi-company ERP. One installation serves several companies; every business record belongs to exactly one company and users only see their own company's data (Super Admin sees all). It covers HR (employees, onboarding, attendance, leave, payroll, performance, offboarding), purchasing, accounting (invoices, bills, payments, reports), expenses, tasks/projects, meetings, compliance, documents and templates, policies, support, marketing and website records.

Item	Value
Framework	CodeIgniter 4.7 (PHP 8.1+; developed on 8.2) with a modular (HMVC-style) layout
Database	MySQL 8 locally (`erp_db` , port 3307); MariaDB on the live host (`u160045728_sayeon`)
PDF	Dompdf (generated documents, invoices, payslips, reports)
Spreadsheet	PhpSpreadsheet
UI	Server-rendered PHP views, Bootstrap/AdminLTE style, `public/assets/css/sayeon-theme.css`
Live site	https://sayeon.sanvima.net/

A2. Repository layout

```
app/
  Config/            Routes.php (includes every module's Routes.php), Filters.php, Events.php, Services.php …
  Controllers/       BaseController (company scoping, activity log, templated documents), FileController (gated downloads)
  Helpers/           permission_helper (can()), gst_helper, payroll_helper, localization_helper, company_code_helper
  Libraries/         FileStorage (local / S3 adapters), FormBreadcrumbs
  Commands/          CLI jobs (see section A9)
  Database/          core migrations (roles, permissions, users, notifications, activity logs) and seeders
  Modules/<Name>/    Controllers, Models, Services, Views, Routes.php, Database/Migrations (+ Seeds)
  Views/layouts/     main.php (sidebar, topbar, flash messages)
public/              index.php (front controller), assets/
writable/            logs, cache, session, uploads (company_documents, document_templates, generated_documents …)
tests/_support/      E2E harness (base class, bootstrap, phpunit.phar)
tests/e2e/           End-to-end workflow tests
docs/                this documentation
```

Modules

Module	Purpose	Main tables
Auth	Login, profile, users, roles, localization settings	users, roles, permissions, role_permissions, app_settings
Company	Companies, directors, bank accounts, countries/states	companies, company_directors, bank_accounts, countries, states
Department	Departments	departments
Customer	Customers per company	customers
HR	Employees, attendance, leave, payroll, performance, increment letters	employee_profiles, attendance, leave_types, leave_requests, salary_structures, payroll_runs, payslips, review_cycles, performance_reviews, employee_increment_letters
Onboarding	Recruitment pipeline, candidate portal, role-specific candidate form + HR evaluation, onboarding checklist	onboarding_records, onboarding_tasks, onboarding_phases, onboarding_candidate_*, onboarding_role_groups/questions/eval_criteria, onboarding_evaluations
Offboarding	Resignation/termination, exit checklist	offboarding_records, offboarding_tasks, offboarding_phases
Task	Projects and tasks (checklists, comments, attachments, status history)	projects, tasks, task_*
Meeting	Meetings, participants, minutes, action items, recurring meetings, Google Meet / Zoom	meetings, meeting_participants, meeting_action_items, recurring_meetings, google_oauth_connections

Module	Purpose	Main tables
Compliance	Compliance items, recurring cycles, per-company applicable types	compliance_items, compliance_types, company_compliance_types, compliance_attachments
Purchase	Vendors, purchase requests/orders	vendors, purchase_orders, purchase_order_items
Accounting	Invoices (GST), bills, payments	invoices, invoice_items, bills, bill_items, payments
Expense	Subscriptions/expenses, recurring expenses, **expense claims**	expenses, expense_categories, recurring_expenses, expense_claims
Document	Company document store, **template-based document generation** and archive	documents, document_templates, generated_documents
Report	Task, compliance, company, expense, receivables, payables, financial, P&L, income-vs-expense, campaign reports	– (reads other modules)
Notification	In-app notifications, due-date reminders, mail service	notifications
Dashboard / Calendar / Search	Role dashboards, calendar feed, live search	–
Policy, Support, Todo, Marketing, Website	Policies, support tickets, personal to-dos, campaigns, website/domain records	policies, support_tickets, todos, campaigns, websites

A3. Roles, permissions and company isolation

A3.1 Roles

`super_admin`, `admin` (Company Admin), `manager` (Project Manager), `hr` (HR Manager), `accountant`, `compliance_officer`, `employee`.

A3.2 How access is enforced (three layers)

1. **Login filter** (`auth`) on every route except login, assets and the candidate portal.
2. **Route permission filter** – each route declares `permission:<module>.<action>`; the user's permissions are loaded into the session at login. Super Admin bypasses this check.
3. **Controller rules** – company scope (`outOfScope()` / `companyScopeFor()` in `BaseController`), ownership rules (nobody approves their own request), and role rules (e.g. only HR/Super Admin move onboarding stages).

A record outside the viewer's company behaves as *not found*. A company-scoped account that has no employee profile yet is scoped to "no company" and sees **nothing** (not everything).

A3.3 Permission matrix (V = view, C = create, E = edit, D = delete, A = approve, R = reimburse)

Module	Company Admin	Manager	HR	Accountant	Compliance	Employee
attendance	CV	–	CEV	–	–	CV
leave	ACV	–	ACEV	–	–	CV
payroll	CEV	–	CEV	–	–	V (own payslips)
employee	CDEV	–	CDEV	–	–	–
onboarding	EV	EV	CDEV	EV	–	V (own)
offboarding	EV	EV	CDEV	EV	–	CEV (own exit)
task	CDEV	CDEV	–	–	–	CEV
project	CEV	CDEV	–	–	–	V
meeting	CDEV	CDEV	–	–	–	CEV
compliance	CEV	–	–	–	CDEV	–
purchase_order	ACDEV	CV	–	ACV	–	CV

Module	Company Admin	Manager	HR	Accountant	Compliance	Employee
invoice / bill / customer	CDEV	–	–	CDEV	–	–
expense	CDEV	–	–	CDEV	–	–
expense_claim	ACRV	ACV	ACV	ACRV	CV	CV
document / policy	CDEV	–	–	–	CDEV	V
bank_account	CDEV	–	–	–	–	–
report	V	V	V	V	V	–

Super Admin has every permission plus user/role/company/setting management and the "All companies / one company" switcher. The authoritative source is the `role_permissions` table (seeded by `RolePermissionSeeder`, extended by migrations).

Notes that surprise people: * **Managers hold no leave permissions** (by design); leave approval belongs to HR / Company Admin. * **Only HR and Super Admin move onboarding/offboarding stages** even though Company Admin/Manager/Accountant hold `.edit` (they may complete their own checklist items). * Leave approval hierarchy: Super Admin > Company Admin > HR = Manager > Accountant / Compliance / Employee. An approver may only decide requests from a strictly lower rank, in their own company; nobody approves their own.

A4. Database

- Migrations live in `app/Database/Migrations` and in each module's `Database/Migrations`. Run with `php spark migrate -all`.
- Every company-owned table carries `company_id` (or reaches a company through `employee_profiles`).
- Money columns are `DECIMAL(12,2)`; dates are `DATE` / `DATETIME` in the application timezone (`Asia/Kolkata`).
- **Pending migrations for live** (created 13 Oct 2026, present only on the test database until you run them): `HR/2026-10-13-100001_AddCompanyToPayrollRuns`, `Expense/2026-10-13-110001_CreateExpenseClaimsTable`, `Accounting/2026-10-13-120001_AddPurchaseOrderToBills`. Code for payroll runs, expense claims and bill↔PO links requires them.

A5. Cross-cutting behaviours

Concern	How it works
Activity log	`logActivity(module, action, recordId, description)` writes `activity_logs` after creates/updates/deletes.
Notifications	`NotificationService::notify()` writes in-app rows (`notifications`). Reminders are produced by a sweep run from the navbar bell (`GET /notifications/recent`, at most every 5 minutes per session) for tasks, compliance items, today's meetings, expense renewals and website expiries; one reminder per item per type per day.
Email	`MailService` / onboarding mail send only when SMTP host, user and password are all configured; otherwise the app reports "not configured" and shows the link/message instead. Nothing is ever claimed as sent when it was not.
Files	`service('fileStorage')` (local `writable/uploads` or S3). Downloads go through `files/download`, which re-checks permissions.
Documents from templates	HR/Offboarding/Payroll documents (offer, joining, relieving, payslip …) are **template-only**: with no active template for the record's company, generation is blocked. Accounting documents (invoice, bill, PO, expense report) fall back to the built-in layout. A company never uses another company's template. Generated files are archived and an unchanged record re-serves the archived file.
Localization	`app_settings` (org country, currency, timezone, locale, formats); users may override timezone/locale.
Numbering	Employee codes, task codes, project codes and invoice numbers use the company short code (`SAY-INV001`, `SAY-T001`); purchase orders are `PO-YYYY-0001` (global sequence).

A6. Business modules in brief (rules that are enforced)

- **Employees** – one profile per login; unique employee code; reporting manager must be in the same company; date of birth not in the future.
- **Attendance** – self check-in/out (once a day each), HR marking (upsert per employee/day), CSV bulk upload (`employee_code,date,status,check_in,check_out`). Statuses: present, absent, half_day, on_leave, holiday, week_off. "Late" = check-in after 10:00; "early" = check-out before 17:00.
- **Leave** – pending → approved / rejected / cancelled. Dates must not overlap another pending/approved request; paid leave types are limited by `annual_quota` (pending + approved + new); approving marks each day `on_leave` in attendance.

- **Payroll** – per company and month; Loss of Pay = absent days + approved unpaid-leave days; gross = basic + HRA + allowances; net = gross − deductions − LOP (`gross ÷ days-in-month × LOP days`). Marking a run paid books one cash-out payment per company.
- **Invoices / bills** – server-side totals (discount, CGST+SGST for same-state, IGST for different state, derived from company vs customer state), unique numbers, payments capped at the open balance, no payments on cancelled documents, documents with payments cannot be deleted (cancel instead), status re-evaluated when the total is edited.
- **Reports** – outstanding receivables/payables are `amount − payments`; cash figures come from the `payments` ledger (invoice receipts, bill payments, salary runs, expense-claim reimbursements).
- **Tasks** – every task needs a company, department and a project of the same company; assignee must be in the company; status history is recorded; `completed_at` follows the status.
- **Meetings** – provider `none` (manual), Google Meet or Zoom; participants must belong to the company; employees reach only meetings they organise or are invited to; join links use an unguessable token.
- **Compliance** – items per company/type with due date, recurrence (none, monthly, quarterly, half-yearly, annually) and reminder lead time; filing spawns the next cycle exactly once; a daily generator also pre-creates upcoming cycles.
- **Expenses** – subscription/renewal expenses with renewal reminders; recurring templates generate expense rows idempotently; separate **expense claims** (see workflows doc).

A7. Setup

Local development

1. PHP 8.2 with `mysqli`, `intl`, `mbstring`, `gd`, `zip`, `curl`; MySQL 8.
2. Copy `.env.example` to `.env`; set `database.default.*`, `app.baseURL`, mail and provider keys.
3. `php spark migrate -all`, then seeders (`RoleSeeder`, `PermissionSeeder`, `RolePermissionSeeder`, `CompanySeeder`, …) as needed.
4. `start-dev.bat` starts MySQL (3307), PHP-CGI workers and nginx on `http://127.0.0.1:8080`.
5. Test logins are listed in `login.txt` (shared password for test users). **Change all default passwords before going live.**

Environment (.env) keys that matter

`CI_ENVIRONMENT`, `app.baseURL`, `database.default.*`, `encryption.key` (do **not** change on a system that already stores encrypted values – bank account numbers and website credentials), `storage.driver`, `email.*`, `meetingProviders.*` (Google/Zoom), `whatsapp.*`, `app.superAdminEmail`.

A8. Deployment to the live site

Summary (the full step-by-step with the exact upload list was prepared separately in `C:\Users\mamat\sayeon_deploy` and `…_SENSITIVE`): 1. Back up live files and the live database. 2. Upload `app/`, `vendor/`, `public/` (including `.htaccess`), `spark`, composer files and empty `writable/` folders. **Never** upload the local `.env`, `.git`, `tests/`, `writable` contents, SQL dumps or documents. 3. Create the server `.env` from `.env.production` (base URL `https://sayeon.sanvima.net/`, live DB password, keep the existing `encryption.key`). 4. Run the database upgrade SQL (structure first, then reference data) in phpMyAdmin, or `php spark migrate -all` over SSH. 5. Make `writable/` writable; upload template files / documents that belong to synced data. 6. Smoke-test login, an employee list, an invoice PDF, a payslip PDF and an upload; check `writable/logs`.

A9. Scheduled work

Job	Command	Also runs automatically
Next compliance cycles	`php spark compliance:generate-recurring`	once a day on the first request after midnight (Events.php)
Recurring expenses	`php spark recurring-expenses:generate`	same
Recurring meetings	`php spark recurring-meetings:generate`	same
Template self-check	`php spark documents:selftest`	no (diagnostic; rolls back everything it creates)

On the live host add a daily cron calling the three generate commands so nothing depends on someone opening the site.

A10. Testing

- **Harness**: `tests/_support/E2ETestCase.php` runs real requests (routes → filters → controllers → models → MySQL) against the throw-away database `erp_e2e_test` (built from `erp_db`'s schema by `build_test_db.php`). It refuses to run on any other database, blanks SMTP credentials (no mail is sent) and empties business tables before every test.
- **Runner**: PHPUnit 10.5 phar in `tests/_support/`. Run from the project root with a scratch `phpunit.xml` whose bootstrap is `tests/_support/bootstrap.php`, one test class per PHP process.
- **Suites** (`tests/e2e`): recruitment→employee, attendance/leave, payroll/payslip, offboarding, procurement→payable, invoice→cash, expense subscriptions, tasks, meeting→task, compliance, document generation, access control and isolation, meeting providers, candidate profile.
- **Known-defect convention**: a test for a rule the code does not yet enforce reports as *incomplete* with a defect id instead of failing; it turns green automatically when fixed.
- **Environment notes (Windows)**: an antivirus may quarantine test files run from inside the project folder – run from a scratch copy. The tests replace the file-storage adapter with a copy-based one because renaming a freshly written PDF can be refused by the scanner.

A11. Known gaps and limitations (open items)

Id	Area	Gap
D-13	Purchase	A purchase order with no linked bill still has a manually set payment status.
D-22	Tasks	Employees only *list* their own tasks, but can open/update any task of their company by id.
–	Expenses	Claims have no receipt upload and do not feed the profit-and-loss statement (cash ledger only).
–	Accounting	No double-entry ledger or chart of accounts: "accounting entries" are invoices, bills and the payments ledger. Payroll cost is not part of the P&L.
–	Invoices	No approval step; credit notes / advances are not modelled.
–	Procurement	No separate goods-receipt document; "received" is a status on the order.
–	Meetings	No "completed" state; a meeting is a dated record with minutes.
–	Compliance	No reviewer/approval step; "in progress" cannot be set from the UI.
–	Offboarding	Final settlement is a checklist item only (no amount calculation).

Defects found and fixed during the 2026-10 review are listed in Part B, section B14.

Part B – End-to-end business workflows

Everything below was verified against the code and exercised by the automated suite in `tests/e2e`. Where a step you might expect does **not** exist, it is marked **Not supported**.

B0. Conventions

- **Role** names: SA = Super Admin, CA = Company Admin, MGR = Manager, HR, ACC = Accountant, CO = Compliance Officer, EMP = Employee.
- **Company isolation** applies to every flow: a record outside the viewer's company is "not found". Cross-company actions (assigning, approving, linking) are refused.
- **Idempotency**: repeating a click/POST never creates a second record or a second payment (specific protections are listed per flow).
- **Notifications** are in-app (`notifications` table). Email is only sent if SMTP is configured; otherwise the screen says so.
- Every create/update/delete writes `activity_logs`.

B1. Module dependency map

From	Hands over to	What flows across
Recruitment / Onboarding	User + Employee profile	Candidate details become the login (role Employee) and the employee record (company code, manager, designation, joining date)
Employee profile	Attendance, Leave, Payroll, Offboarding	Everything HR does for a person hangs off the profile and its company
Attendance	Payroll	Absent days become Loss of Pay
Leave	Attendance, Payroll	Approved leave is written to attendance as `on_leave`; approved unpaid leave days become Loss of Pay
Payroll run	Payslip document, Payments ledger	Payslip PDF from the company template; marking the run paid books the salary payment
Offboarding	Employee profile, User, Payroll	Exit completed: status resigned/terminated, login disabled, employee left out of later payroll
Purchase order	Bill	An approved order can be billed; its payment status then follows the bill
Bill / Invoice	Payments ledger	Payments out (bills) and in (invoices)
Expense claim	Payments ledger	Reimbursement books a cash-out payment
Payments ledger, invoices, bills	Reports	Receivables, payables, cash flow, financial summary, P&L
Meeting action item	Task	Conversion creates the task; completing the task completes the item
Compliance item	Next compliance cycle	Filing (or the daily generator) creates the next period once
Document template (per company and type)	Generated document archive	Offer, joining, relieving, payslip, invoice and similar documents; archived and immutable once issued

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B2. Employee lifecycle: Recruitment → Onboarding → Employee → Attendance → Leave → Payroll → Offboarding

In the application the employee record is created **by** onboarding (when a candidate has joined), so the order is *Onboarding* → *Employee* → *Attendance* → *Payroll* → *Offboarding*. An employee can also be created directly by HR for an existing login.

B2.1 Recruitment and onboarding

Tests: `RecruitmentToEmployeeFlowTest`, `OnboardingCandidateProfileFlowTest`

Starting conditions: company, department, a reporting manager who has an employee profile in the same company, an HR user. The candidate has no ERP login. **Owners:** HR and Super Admin only (Company Admin, Manager and Accountant can view and complete their own checklist items).

#	Action (role)	Route	Status change	Side effects
1	Add candidate (HR) – name, email, phone, source, department, designation, reporting manager	`POST hr/onboarding`	→ `candidate`	Manager notified (`onboarding_assigned`)
2	Select candidate (HR)	`POST hr/onboarding/{id}/select`	`candidate` → `selected_for_onboarding`	–
3	Send portal link (HR)	`…/send-portal-link`	→ `link_sent` , verification `pending`	48-char token valid 7 days; mailed only if SMTP configured, otherwise the link is shown on screen. A new link invalidates the old one. After the offer is sent HR can re-issue the link without changing the stage
4	Candidate submits details (public portal, token only)	`POST onboarding/portal/{id}/{token}/verify`	→ `details_submitted` , verification `submitted`	A second submit is refused until HR asks for a correction. Role-specific form/answers are stored in the `onboarding_candidate_*` tables
5	HR starts review	`…/start-review`	→ `hr_review`	–
6	HR decision	`…/verification/review` (`approved` / `rejected` / `correction_needed`)	`approved` → **approved** (verification `verified`); `rejected` → **rejected** (closed); `correction_needed` → back to `link_sent` with HR's note	Reviewer and time recorded
7	Send offer (HR)	`…/send-offer`	`approved` → `offer_sent`	`offer_sent_at` ; offer letter reachable by the candidate through the token – **requires an active Offer Letter template for the company**
8	Candidate responds	`POST onboarding/portal/{id}/{token}/offer-response`	`accepted` → `offer_accepted` (default onboarding checklist created once); `rejected` → `withdrawn`	Only possible after HR approval and while the token is valid
9	HR advances the pre-joining stages	`…/status` (one defined next step at a time)	`offer_accepted` → `documents_pending` → `bgv` → `hr_verification` → `ready_to_join` → `joining` → `joined`	Skipping a stage is refused; timestamps (`hr_verified_at` , `joined_at`) set
10	Create employee record (HR)	`…/create-employee` (allowed from `joined` , `induction` , `probation` , `confirmed`)	–	One transaction: user (role Employee, temporary password shown once) + employee profile (company-prefixed code, manager, designation, joining date) + link to the onboarding record. Existing login in the same company is linked; in another company is refused; repeating is a no-op
11	Post-joining	`…/status`	`joined` → `induction` → `probation` → `confirmed`	Manager notified of each stage
–	Withdraw (HR)	`…/withdraw`	any open stage → `withdrawn` (not from `rejected` , `withdrawn` , `confirmed`)	–

Rejection / cancellation paths: HR rejects (→ `rejected`), HR requests correction (loop back to step 3's state), candidate declines (→ `withdrawn`), HR withdraws. A rejected or withdrawn candidate cannot be offered, advanced or turned into an employee. **Errors covered:** invalid email/name,

wrong/expired token, offer response before approval, manager from another company, other company's HR, roles without permission.

B2.2 Employee record

Tests: `EmployeeAttendanceLeaveFlowTest`

HR/Company Admin create a profile for an existing login (`POST hr/employees`): unique login and unique employee code, company in the creator's scope, reporting manager in the **same company**, DOB not in the future. Employees, Managers, Accountants and Compliance cannot create. Lists show only the viewer's company.

B2.3 Attendance

- Employee self check-in / check-out (`hr/attendance/check-in|check-out`): check-in once a day, check-out only after check-in and once. Super Admin has no self-service.
- HR marks/corrects a day (`hr/attendance/mark`) – one row per employee per day, own company only, status from: present, absent, half_day, on_leave, holiday, week_off.
- HR CSV bulk upload (`hr/attendance/bulk-upload`) – unknown/other-company employee codes and bad rows are skipped and counted.
- Attendance feeds Payroll (absent days) and the employee profile/dashboard KPIs.

B2.4 Leave

Step	Rule
Apply (EMP, HR, CA; **not** Manager/Super Admin)	Valid leave type; end ≥ start; days = calendar days inclusive; **no overlap** with a pending/approved request; **paid types must fit the annual quota** (pending + approved + this request); unpaid leave is not capped. Reporting manager is notified
Decide (`approve` / `reject`)	Only a pending request; approver must have `leave.approve`, be in the same company, outrank the requester (SA > CA > HR = MGR > ACC/CO/EMP) and not be the requester (Super Admin excepted). Requester is notified once
Result of approval	Every day of the request is written to attendance as `on_leave` (a day HR had marked `absent` is corrected; present / half-day / holiday / week-off rows are kept)
Cancel	Only the owner, only while pending
Statuses: `pending` → `approved` | `rejected` | `cancelled`.	

B2.5 Payroll and payslip

Tests: `PayrollPayslipFlowTest`

1. **Salary structure** (`POST hr/payroll/structure/{userId}` – HR, CA): basic, HRA, allowances, deductions; saving again updates the single row; own company only.
2. **Generate** (`POST hr/payroll/generate`, month 1–12, year 2000–2100 – HR, CA; Super Admin creates one run per company): * payable = active employees of the company with a salary structure; * `gross = basic + HRA + allowances`; **LOP days = absent attendance days + approved unpaid-leave days** in the month; `LOP amount = gross ÷ days-in-month × LOP days`; `net = gross − deductions − LOP`; * one run per **company + month + year** (a repeat is refused: "already exists"); one payslip per employee; every employee notified once; all in one transaction.
3. **Payslip view/PDF**: the employee sees their own; HR/CA see their company's; others and other companies are refused. The PDF **requires an active Payslip template**; generated once and re-served while the data is unchanged.
4. **Mark paid** (`POST hr/payroll/{id}/mark-paid` – HR, CA): run `processed` → `paid`, and **one cash-out payment per company** (method `payroll`, reference `PAYROLL-YYYY-MM`) is booked in the payments ledger. Repeating changes nothing. Salary payments cannot be deleted from the invoice/bill screens. Another company cannot mark the run paid. Statuses: `processed` → `paid`. Resigned/terminated employees are not paid.

B2.6 Offboarding

Tests: `OffboardingExitFlowTest`

Step	Detail
Start	EMP files a resignation for themself (`POST hr/offboarding`, type forced to *resignation*, only one open exit); HR files a resignation or termination for any employee of their company. 11 checklist items are created; reporting manager notified

Step	Detail
Checklist (`…/tasks/{id}`)	Each item has an owner role: Manager (review, handover acceptance, department clearance), HR (approval, exit interview, final documents), Employee (handover notes, asset return), Admin (asset confirmation, access revocation), Accountant (final settlement). Only the owner role – or HR/Super Admin – can update an item; statuses `pending` , `in_progress` , `completed` , `skipped` . The record's stage moves forward automatically with the checklist
Stages	`resignation_submitted` → `manager_review` → `hr_approval` → `exit_initiated` → `handover` → `department_clearances` → `asset_return` → `access_revocation` → `final_settlement` → `exit_interview` → `final_documents` → `exit_completed` ; side exit `rescinded`
Close (`…/status` = `exit_completed` , HR/SA only)	Allowed only when **every checklist item is completed or skipped**. Then: employee profile → `resigned` or `terminated` , `users.status` → `inactive` (login disabled), `completed_at` set, employee and manager notified. History (payslips, leave, tasks) is kept
Final states	`exit_completed` and `rescinded` are final – no further stage changes
Documents	Relieving/exit letters need an active template for the company
Not supported	Final settlement is a checklist item only – no amounts, payment or accounting entry

B3. Purchase → Approval → Order → Payment → Accounting

Tests: `ProcurementToPayableFlowTest`

There is **one** purchase document: the purchase order, which starts life as a request (`pending`). There is no separate purchase-request entity and no goods-receipt document.

#	Action (role)	Route	Status change	Side effects / rules
0	Maintain vendors (CA)	`vendors`	–	Per company; a vendor with orders cannot be deleted (set inactive)
1	Request a purchase (EMP, MGR, ACC, CA) – vendor, items (qty × unit price)	`POST purchase-orders`	approval `pending` , payment `pending` , no fulfillment	`PO-YYYY-0001` ; total computed server-side; vendor must be in the same company; invalid/blank items refused
2	Edit while pending (CA)	`…/{id}/edit`	–	Items replaced, not appended; approved/rejected orders cannot be edited
3	Approve or reject (ACC, CA – holders of `purchase_order.approve`)	`…/approve` , `…/reject`	`pending` → `approved` (fulfillment `ordered`) or `rejected`	**Requester cannot decide their own request** (Super Admin excepted); another company cannot; decided once; requester notified
4	Track fulfillment (CA)	`…/status`	`ordered` → `in_transit` → `received`	Only on approved orders; moves forward only (a received order is not taken back)
5	Record the vendor's invoice as a **Bill** (ACC, CA) and link it to the approved order of the same company	`accounting/bills` (field *Purchase order*)	bill `unpaid`	GST totals; bill number unique
6	Pay the bill (ACC, CA)	`accounting/bills/{id}/payments`	bill `unpaid` → `partially_paid` → `paid` (`overdue` set automatically after the due date)	Payment (direction *out*) in the ledger; cannot exceed the open balance or be posted on a cancelled bill

#	Action (role)	Route	Status change	Side effects / rules
7	Payment status follows Accounting	automatic	PO payment `pending` → `partial` → `paid`	Derived from the linked bills; cannot be flagged by hand once a bill is linked
Delete: an approved order can only be deleted while it has not progressed – no linked bills, fulfillment still `ordered`, payment still `pending`. Pending and rejected orders can be deleted by Company Admin.				

Accounting outputs: bill/payables list, payments ledger, Payables, Financial Summary and P&L reports. **Gaps:** payment status of an order with no linked bill is still a manual flag (D-13); no three-way match enforcement.

B4. Task → Assignment → Review → Completion

Tests: `TaskLifecycleFlowTest`

#	Action	Rules
1	Create task (MGR, CA, EMP, SA)	Title, company, department, **project of the same company**, priority (low/medium/high/urgent), optional assignee **in the same company**, dates; code like `SAY-T001`; UUID assigned; status history row written
2	Assignment	Assignee notified (not when assigning yourself); re-assigning notifies only the new assignee
3	Work	`POST tasks/{id}/status`: `new`, `assigned`, `in_progress`, `waiting`, `review`, `completed`, `on_hold`, `cancelled`. Every change is recorded (from → to, who, when); `completed_at` is set on completion and cleared if the task is re-opened
4	Review	`review` is a status only. There is **no reviewer role or approve/reject action** – anyone with task edit rights in the company can move a task to or out of `review` / `completed`
5	Reminders	Open tasks due within 2 days → `task_due_soon`; past due → `task_overdue` (once a day per task). Completed/cancelled tasks never remind
6	Comments, checklist, attachments	Company-scoped; empty comments refused
7	Delete (MGR, CA)	Soft delete; history kept
Also: tasks created from meeting action items (section B5) and from the To-Do list are ordinary tasks.		
Known gap (D-22): employees only *list* their own tasks, but can open or update a colleague's task by id.		

B5. Meeting → Action Item → Task → Completion

Tests: `MeetingToTaskFlowTest`, `MeetingProvidersTest`

#	Action	Rules
1	Schedule (MGR, CA, EMP, SA)	`POST meetings`; provider `none` (no link generated), Google Meet or Zoom (created before saving – a failure saves nothing); end time after start; all participants must be in the company; the organiser is always a participant

#	Action	Rules
2	Invite	Each participant gets an in-app invitation (not the organiser); email only if configured; external participants by name/email, one invite per email; the same user cannot be added twice
3	Minutes	`POST meetings/{id}/mom`
4	Action items	`POST meetings/{id}/action-items`: description, optional assignee **who belongs to the meeting's company**, due date; status `pending` → `in_progress` → `completed` (other values refused)
5	Convert to task (needs `task.create`)	`…/action-items/{item}/convert` with **department and project of the meeting's company**: one transaction creates the task (`Follow-up: …`, same company/assignee/due date, status `new`, history row) and links it to the item; assignee is notified. Converting twice is refused; without a project/department nothing is created
6	Completion	Completing the linked task sets the action item to `completed`; re-opening the task re-opens the item
Visibility: an employee reaches a meeting (view, minutes, participants, action items) only if they organised or were invited; the join link needs the meeting's token. Another company never sees it.		
Not supported: meeting "completed" state (a meeting is a dated record plus minutes).		

B6. Compliance → Reminder → Submission → Completion

Tests: `ComplianceCycleFlowTest`

#	Action (role)	Rules
1	Create item (CO, CA, SA)	Company, compliance type, title, regulator, period label, **due date**, recurrence (none / monthly / quarterly / half-yearly / annually), **responsible person in the same company**, reminder lead days (default 7). Status `pending`; responsible person notified
2	Reminder	Within the lead time: `compliance_due_soon`; after the due date the list flips the item to `overdue` and `compliance_overdue` is sent – once a day, only to the responsible person, never for filed items
3	Submission – "Mark filed" (CO, CA)	`POST compliance/{id}/mark-filed` → status `filed`, `filed_at` stamped; filing an already-filed item keeps the original time
4	Next cycle	Filing a recurring item creates the next cycle once (due date advanced by the recurrence, status `pending`, period label left blank). A daily generator (and *Generate recurring* button) pre-creates cycles for items inside their lead time; filing and the generator never create two successors
5	Completion	`filed` is the end state of a cycle; history chain via `previous_item_id`
Statuses: `pending` → (`overdue` automatically) → `filed`. **Not supported:** reviewer/approval step; `in_progress` cannot be set from the UI.		

B7. Invoice and accounting

Tests: `InvoiceToCashFlowTest`

#	Action	Rules

1	Create invoice (ACC, CA, SA)	Unique number (per-company sequence suggested), customer name + country (+ state), items, discount %, GST %. **Server computes**: subtotal, discount, taxable value, tax. Same state as the selling company → CGST + SGST; different state → IGST (a posted GST type cannot override this); total rounded to 2 decimals; oversize totals refused
2	Status	`draft`, `sent`, `partially_paid`, `paid`, `overdue`, `cancelled`. Payments drive `partially_paid` / `paid`; `overdue` is set automatically for open invoices past their due date. **No approval step**
3	Record payment	Amount > 0 and ≤ open balance; not on cancelled invoices; payment keeps the invoice's company; removing a payment recomputes the status
4	Edit	If the total changes after payments the status is re-evaluated
5	Delete	Refused when payments exist (cancel instead)
6	Documents	PDF/print: built-in layout, or the company's Invoice template when active (archived)
7	Reports	Cash in/out and net cash flow from the payments ledger; outstanding receivables/payables = open `amount − payments`; P&L (accrual, by month); Income vs Expense
Other payments in the same ledger: bill payments (out), salary runs (out), expense-claim reimbursements (out).		

B8. Expenses

Tests: `ExpenseSubscriptionFlowTest`

B8.1 Company expenses / subscriptions

Create/edit/cancel expenses (ACC, CA): vendor, category, billing cycle (monthly/quarterly/yearly/one_time), amount, renewal date, status `active` / `cancelled`. No approval. Renewal reminders (`expense_due_soon` within 7 days, `expense_overdue` after) go to the creator once a day. **Recurring templates** (monthly/quarterly/yearly; end by occurrences or end date) generate expense rows exactly once per occurrence, catching up missed periods; templates can be paused, resumed, cancelled; a run is limited to the viewer's company (Super Admin: all).

B8.2 Expense claims (employee reimbursement)

#	Action (role)	Route	Status	Rules
1	Submit (any role with `expense_claim.create`)	`POST expense-claims`	`submitted`	Title, category, amount > 0, date not in the future; claimant must have an employee profile; company = claimant's; identical waiting claim refused; reporting manager notified
2	Cancel (claimant)	`…/cancel`	→ `cancelled`	Only while `submitted`
3	Approve / reject (MGR, HR, ACC, CA)	`…/approve`, `…/reject` (optional note)	`submitted` → `approved` / `rejected`	Same company; not your own claim; once only; claimant notified
4	Reimburse (ACC, CA)	`…/reimburse`	`approved` → `reimbursed`	One transaction: cash-out payment (reference `CLAIM-{id}`, method as entered) + claim linked to it (`payment_id`); not your own claim; cannot be done twice; claimant notified
Visibility: claimants see their own claims; approvers/Finance see their company's.				
Not supported: receipt upload; posting to the P&L (cash ledger only).				

B9. Document generation

Tests: `DocumentGenerationFlowTest`

1. **Template configuration** (CA, CO with `document.edit`): per company and document type, versioned; file type html / docx / pdf; flagged auto-detected fields must be reviewed; a **draft must be previewed** before activation; activating a version retires the previous one – exactly one active version per company + type.
2. **Data mapping**: placeholders such as `{{employee_name}}`, `{{designation}}`, `{{joining_date}}`, `{{annual_salary}}`, `{{net_pay}}`, `{{period}}` are filled from the ERP record. A template that uses an unknown field, or has no place for the record's identity, is refused – nothing is printed blank.
3. **Generation**: via the ERP action (offer/joining letter, payslip, relieving letter, increment letter, invoice, bill, PO, expense report) or `documents/render/{type}/{id}`. Only the record's own company template is used. Missing required data produces a "complete these details" message.
4. **Download and archive**: the PDF/DOCX is stored (`generated_documents`: template version, data snapshot, hash, user). Unchanged data re-serves the stored file byte-for-byte; changed data or a new template version creates a **new** document and never alters an issued one.
5. **Rules**: HR letters/certificates/payslips are **template-only** (no template → blocked); accounting documents fall back to the built-in layout. Templates that produced documents cannot be deleted; an active template must be deactivated first; other companies cannot see or change them.

B10. Access control and isolation

Tests: `AccessControlAndIsolationTest`

- Login with valid credentials sets role, permissions and (for Super Admin) the "all companies" scope; wrong password and unknown email give the same message; deactivated or deleted users cannot log in.
- Every protected URL (GET and POST) redirects anonymous visitors to the login.
- A role × module matrix test confirms each role reaches exactly the screens its permissions grant.
- User and role administration is Super Admin only; the company switcher is Super Admin only.
- List screens (vendors, expenses, compliance, projects, customers, meetings, invoices, campaigns, websites, employees, tasks) show nothing from other companies – including for a Company Admin account that has no employee profile yet.

B11. Coverage summary

Workflow	Test file	Result at last run
Recruitment & onboarding → employee	`RecruitmentToEmployeeFlowTest`, `OnboardingCandidateProfileFlowTest`	Pass
Employee, attendance, leave	`EmployeeAttendanceLeaveFlowTest`	Pass
Payroll & payslip	`PayrollPayslipFlowTest`	Pass
Offboarding	`OffboardingExitFlowTest`	Pass
Purchase → payment	`ProcurementToPayableFlowTest`	Pass (1 open item: D-13)
Invoice → cash → reports	`InvoiceToCashFlowTest`	Pass
Expenses (subscriptions)	`ExpenseSubscriptionFlowTest`	Pass
Expense claims	verified by a one-off scenario (submit → approve/reject → reimburse → ledger); **no permanent test file yet**	Pass (not in suite)
Task	`TaskLifecycleFlowTest`	Pass (1 open item: D-22)
Meeting → task	`MeetingToTaskFlowTest`, `MeetingProvidersTest`	Pass
Compliance	`ComplianceCycleFlowTest`	Pass
Document generation	`DocumentGenerationFlowTest`	Pass

Workflow	Test file	Result at last run
Access control & isolation	`AccessControlAndIsolationTest`	Pass

B12. Test data and preconditions

`E2ETestCase::world()` builds two companies (Alpha/ALP, Beta/BET) each with Company Admin, Manager (reporting manager of the employee), HR, Accountant, Compliance Officer, Employee, plus a Super Admin and a department; extra helpers create salary structures, projects, templates and logins. Reference data (roles, permissions, countries, states, leave types, compliance types, phases) comes from the database clone. All test users use the password `Test@123`.

B13. Status transition quick reference

Entity	Statuses (→ = allowed move)
Onboarding record	candidate → selected_for_onboarding → link_sent → details_submitted → hr_review → approved → offer_sent → offer_accepted → documents_pending → bgv → hr_verification → ready_to_join → joining → joined → induction → probation → confirmed; side exits: rejected, withdrawn; correction_needed returns to link_sent
Leave	pending → approved | rejected | cancelled
Payroll run	processed → paid
Offboarding	resignation_submitted → … → exit_completed (needs checklist done); side exit rescinded; both final
Purchase order	approval: pending → approved | rejected; fulfillment: ordered → in_transit → received; payment: pending → partial → paid (from bills)
Bill	unpaid → partially_paid → paid (+ overdue, cancelled)
Invoice	draft → sent → partially_paid → paid (+ overdue, cancelled)
Expense claim	submitted → approved → reimbursed; submitted → rejected | cancelled
Task	new, assigned, in_progress, waiting, review, completed, on_hold, cancelled (history kept)
Meeting action item	pending → in_progress → completed
Compliance item	pending → overdue → filed
Recurring expense/meeting	active ⇄ paused; active → completed | cancelled

B14. Defects found and fixed

Found by the end-to-end review (October 2026) and fixed in the code:

Area	Problem fixed
Payroll	Payroll was global: one company's run for a month blocked all others. Now per company (migration). Any company could mark another's run paid – now scoped. Marking paid now books the salary payment
Accounting	Overpayment, payments on cancelled documents, deleting invoices/bills with payments (which silently deleted the payments), stale status after editing a paid invoice, receivables overstated by part-paid invoices
Meetings → tasks	Converting an action item to a task always failed (no project); assignee not notified; action item did not follow task completion; unvalidated status/assignee; duplicate participant crashed; uninvited employees could edit a meeting they could not open
Offboarding	Exit could be closed with open checklist items, a closed exit could be re-opened, the leaver kept a working login
Leave	No overlap check, no quota check, unknown leave type crashed, approved leave did not appear in attendance
Employees / compliance	Reporting manager or responsible person from another company accepted; a filed item's filing time could be overwritten
Tasks	Re-opening a task left the completion date
Isolation	A Company Admin without an employee profile saw projects, invoices, campaigns and websites of every company
Purchase	Orders could be un-received, deleted after payment, or flagged paid independent of Accounting (now linked to bills)

Area	Problem fixed
Onboarding	An expired offer link could not be re-issued

Open items are listed in section A11 of Part A.